

UDP CLIENT-SERVER COMMUNICATION IN PYTHON

AIM:

Implement UDP client-server communication using socket programming in Python.

Server Algorithm:

1. Create socket → `socket.socket()`
2. Bind to IP and port → `bind()`
3. Receive message → `recvfrom()`
4. Send response → `sendto()`
5. Close connection

Client Algorithm:

1. Create socket → `socket.socket()`
2. Send message → `sendto()`
3. Receive response → `recvfrom()`
4. Close connection

CODE:

Server Code:

```
import socket

sockfd = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
print('UDP socket created')
sockfd.bind(('localhost', 55555))
print('Waiting for messages')

while True:
    data, addr = sockfd.recvfrom(1024)
    receivedMsg = data.decode()
    print("Received message from", addr)
    print("Message:", receivedMsg)

    sockfd.sendto(data, addr)
```

```
print("Message reply sent to client")
```

```
choice = input("Do you want to continue (y or n): ")  
if choice.lower() == 'n':  
    break
```

```
sockfd.close()
```

Client Code:

```
import socket
```

```
clientfd = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)  
server_address = ('localhost', 55555)
```

```
name = input("Enter your message: ")  
clientfd.sendto(name.encode(), server_address)
```

```
data, _ = clientfd.recvfrom(1024)  
print("Message received from server:", data.decode())
```

```
clientfd.close()
```

OUTPUT:

<pre>C:\Users\Lenovo\Desktop>python Server.py UDP socket created Waiting for messages Received message from ('127.0.0.1', 65349) Message: Hello Message reply sent to client Do you want to continue (y or n): n C:\Users\Lenovo\Desktop></pre>	<pre>C:\Windows\System32>cd C:\Users\Lenovo\Desktop C:\Users\Lenovo\Desktop>python Client.py Enter your message: Hello Message received from server: Hello C:\Users\Lenovo\Desktop></pre>
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RESULT:

UDP client-server communication was successfully implemented using Python.